TECHNICAL DESIGN DOCUMENT

Loomline: order control

A focused, working slice for Company A, Tiruppur - knitwear exports to UK and EU buyers.

GitHub: https://github.com/aggarwaltanmay/loomline-order-control

Live app: https://loomline-order-control.tanmay-aggarwall.chatgpt.site

01 / Decision and scope

Ship six connected screens: sign-in, order board, order detail with TNA and run-rate maths, floor update, new order, and read-only buyer view. The three supplied POs seed the database. Merchandisers can adjust milestone dates with an audit reason and confirm a buyer-facing warning. Supervisors enter daily quantities, photos, and Tamil or English issue notes.

Why this slice: it closes the information gap behind late discovery. The board turns floor output into a visible warning while there is still time to act. The buyer view reduces routine status email without exposing another buyer's data.

02 / Architecture

Surface	Responsibility
React / Vinext client	Responsive views; IndexedDB queue for pending floor updates; service worker caches the shell only.
Worker API routes	Input validation, role checks, projection assembly, sign-in sessions, CSV export and AI orchestration.
D1 database	Users, sessions, orders, milestones, floor updates and append-only date changes.
R2 bucket	Floor photos; every read rechecks the parent order's buyer or unit scope.
OpenAI Responses API	Optional live classification and English summary for issue notes; server-only key.

Request path: browser -> signed session cookie -> Worker role check -> buyer/unit-scoped D1 query -> authorized R2 photo read. OpenAI is called only by the Worker after a valid floor update.

03 / Data model

Record	Key fields and relationship
users / sessions	Email, role, buyer_id or unit_id, PBKDF2 password hash; hashed session token and expiry.
orders	PO, buyer, style, colour, size JSON, quantity, ex-factory date, unit, current stage, stage_done, baseline_daily.
milestones	Order ID, stage name, due date and sequence; generated from a TNA template.
updates	Idempotent client ID, order/stage, daily quantity/date, original note, category, English summary, photo key, actor.
date_changes	Order, field, old/new dates, reason, actor and timestamp; inserts only, never overwritten.

04 / Main flows

Actor	Flow and control
Merchandiser	Create PO with colour and size quantities -> template milestones -> filtered board -> inspect visible maths -> correct note category -> confirm buyer warning -> audit date edits.
Supervisor	Sign in to assigned unit -> select PO -> enter pieces, note or photo -> save; failed network sends the same UUID from IndexedDB when online.
Buyer	Sign in to assigned buyer -> view only that buyer's orders, milestones, photos, date history and confirmed warning -> export own CSV.

05 / Delay projection - explicit arithmetic

The target is the due date of the current stage in the TNA plan. This exposes a recoverable problem before the final ex-factory date. For today in India time:

Quantity	Calculation
Remaining	max(0, order quantity - current-stage pieces done)
Days left	max(0, target date - today), in calendar days
Needed per day	ceil(remaining / days left); undefined when time is up
Made per day	Recent stage pieces divided by calendar days since the first update in the last 7 days. Before any real update, show the labelled sample baseline.
Signal	Green at or above needed; amber at 85-99%; red below 85% or overdue. Warning flag when amber/red and within 14 days.

Seeded example: B-PO-88240 has 8,000 pieces, 5,200 finished and a 7 Nov finishing milestone. On 2 Nov, 2,800 / 5 days = 560 needed per day. The assumed sample baseline is 390 per day, so the order is red; at that pace finishing is projected for 10 Nov. After a real floor update, the sample baseline is replaced by the actual recent rate.

06 / AI, uncertainty and correction

For a non-empty issue note, the Worker requests strict JSON from OpenAI gpt-6-luna: one of seven English reason categories, one concise English summary, and high/medium/low confidence. The note is treated as data and is limited to 1,000 characters. The original Tamil note remains beside the suggestion. The merchandiser can overwrite the category; the update is marked reviewed.

If the API is unavailable or the daily demo call limit is reached, an explicit Tamil/English keyword rule provides a Rule suggestion or Needs review. It does not pretend to translate. Buyers see neither raw notes nor unconfirmed warnings. A merchandiser must confirm the warning; any new floor update or date edit clears that confirmation.

Verification on 2 Nov 2026: the selected API project returned HTTP 429 because it had no credits. Live model output remains to be retested after funding. The rule fallback and correction flow are working and visible.

07 / Security and resilience

• Public demo uses invented data and shared reviewer passwords. Passwords are PBKDF2 hashed; sessions use hashed tokens in D1 and Secure, HttpOnly, SameSite=Lax cookies.

• Every order, write, export and photo request checks the server-side role and buyer/unit scope. Buyer B cannot ask for Buyer C data by changing a URL.

• Daily updates have client UUIDs and INSERT OR IGNORE, so reconnection retries do not double-count pieces. Photos are resized on the phone before R2 upload.

• The offline queue is IndexedDB and retries while the app is open. The service worker never caches role-scoped API responses.

08 / Interface and delivery

Design decisions

Screen	Reason for its shape
Board	Four small metrics and colour-coded order rows show trouble first. Buyer, unit, ship week, risk and PO filters stay close to the list.
Order detail	The four numbers behind the warning are prominent. TNA, floor log and date history follow in working order.
Supervisor phone form	One order selector, large numeric input, note/voice, photo and one send action; Tamil labels are the default for this role.
Buyer view	Milestones and photos only; no editing controls or internal issue notes. Confirmed warnings and schedule changes remain visible.

Verification

Eight automated checks pass locally: projection thresholds and zero-day edge cases, Tamil rule classification and uncertainty, buyer read/write isolation, supervisor unit isolation, order creation, and date-change audit. TypeScript checks and the production build pass. Browser checks covered sign-in, board, run-rate detail, Tamil note submission, buyer isolation and a 390px phone layout.

Assumptions - deliberately labelled

A1. “Loomline” is a working product name. Company A and buyer labels are fictional placeholders.

A2. The three sample POs have the quantities, styles, dates and current progress supplied in the brief. Unit assignments, size splits, current-stage milestone dates and baseline daily rates (620, 390, 540) are invented only to make the sample usable.

A3. The TNA template uses cutting, printing, stitching, washing, finishing, packing and ex-factory with offsets of -18, -15, -11, -8, -5, -2 and 0 days for new orders; merchandisers may change dates with a reason.

A4. Green/amber/red uses 100% and 85% of needed daily output; the brief did not set thresholds. Calendar days are used in the demo, not production working-day calendars.

A5. Demo supervisor is assigned Unit 2. Reviewer identities, one shared demo password, the seven issue categories, a 20-note/day AI cap and 1,200px photo resizing are implementation assumptions.

A6. Buyer warning confirmation is a merchandiser action and is cleared by new operational facts. “Weekly status export” is an on-demand Excel-compatible CSV; scheduling and PDF/XLSX generation are deferred.

Cuts and next steps

Cuts: costing, invoices, export documents, fabric stock, attendance, payroll, outside messaging and automatic buyer alerts. The sample does not implement closed-browser background sync, dedicated voice transcription, or real user invitation. Next: replace shared demo credentials with individual SSO/invitations, add working-day calendars and per-stage capacity, run a unit pilot, connect buyer accounts, add true XLSX/PDF reports and monitor the brief's two-season shipment and status-email measures.

Operating constraint: the client's INR 7 lakh / six-week ceiling argues for this focused pilot before integration work. The app does not claim the stated 8% to 5% late-shipment improvement yet.